# 6DOF Arm — Wiring Bench Sheet

ESP32 + PCA9685 + 6× MG996R · print & keep at the bench · full guide: hardware/wiring.md

## SAFETY RULES — read before plugging anything in

**1. NEVER put 12 V on the PCA9685 V+ terminal or any servo.** MG996Rs are rated 4.8–7.2 V. The 12 V supply feeds ONLY the converter input.

**2. USB power = single-servo bench tests only.** Full-arm motion requires the 6 V/10 A rail.

**3. Verify converter output ≈ 6.0 V with a meter BEFORE first connection.** Re-check polarity at V+ — reverse polarity kills the PCA9685 instantly.

**4. Converter → PCA9685 wires short and thick** (14–16 AWG). Stalls can pull 15 A+.

**5. Power-up order:** ESP32 USB first (servos stay idle), then the 12 V supply. Reverse to power down.

**6. Pinch points:** on link loss the arm HOLDS (won't drop, won't yield). Kill 12 V before handling.

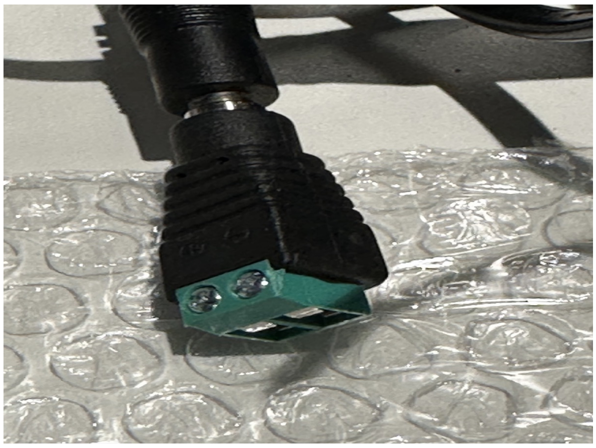
## Servo power path

12 V/10 A supply → buck converter (12–40 V in, 6 V/10 A out) → PCA9685 V+ terminal

## Converter wire colors — straight from the unit's label

| Wire | Role | Connects to |
|---|---|---|
| **Red** | Input + | 12 V supply positive |
| **Black** (paired w/ red) | Input – | 12 V supply negative |
| **Yellow** | Output + (6 V) | PCA9685 **V+** screw terminal |
| **Black** (paired w/ yellow) | Output – | PCA9685 **GND** screw terminal |

The two blacks are per-bundle negatives — use each with its own side. Red side drinks 12 V; yellow side feeds the servos. The PCA9685's onboard 1000 µF cap covers the rail; no extra cap needed.



Left: the converter's own label (IN red/black, OUT yellow/black). Right: barrel-jack → screw-terminal adapter for the 12 V input — mind the +/– marks.

## Logic side: ESP32 ↔ PCA9685 (I2C)

Four female-female jumpers to the PCA9685's 6-pin I2C header (either short end). I2C address 0x40, jumpers open. Board pull-ups are to VCC — which is why VCC must be 3.3 V.

| ESP32 pin | PCA9685 pin | Purpose | Notes |
|---|---|---|---|
| 3V3 | VCC | Logic supply | **3.3 V only** — never the 5 V/VIN pin |
| GPIO21 | SDA | I2C data | Arduino Wire default |
| GPIO22 | SCL | I2C clock | Arduino Wire default |
| GND | GND | Ground | **Required** — common-ground jumper; I2C is dead/flaky without it |
| — | OE | Output enable | Leave unconnected (pulled low = enabled) |

## Servo plugs and channels

Column pin order from board edge inward: **GND (bottom), V+ (middle), PWM (top)**. MG996R leads: **brown = GND, red = V+, orange = signal** — brown faces the board edge.

| Channel | Joint | Position on arm |
|---|---|---|
| 0 | joint1 | Base rotation (yaw) |
| 1 | joint2 | Shoulder pitch |
| 2 | joint3 | Elbow pitch |
| 3 | joint4 | Wrist pitch |
| 4 | joint5 | Wrist roll |
| 5 | joint6 | Gripper |
| 6–15 | — | Spare |

Wrong channel? Move the plug, don't edit firmware — keeps this table true.

## Common ground — required, not optional

PSU(–) → converter IN(–) → converter OUT(–) → PCA9685 GND → ESP32 GND (via the I2C ground jumper). Beep-test all four points.

## Bench bring-up checklist

- [ ] All grounds common (continuity-beep PSU–, converter–, PCA9685 GND, ESP32 GND)
- [ ] Converter output ≈ 6.0 V, verified with meter, no load
- [ ] I2C jumpers: 3V3→VCC · GPIO21→SDA · GPIO22→SCL · GND→GND
- [ ] ESP32 on USB: serial prints “PCA9685 initialized, all channels off”
- [ ] One servo on channel 0 — single-servo sweep (USB/bench power OK)
- [ ] Remaining servos on channels 1–5 per table
- [ ] Full-arm motion ONLY on converter power